



Society of Petroleum Engineers

SPE-226974-MS

Systematic Strategies to Conquer the Technical Challenges to Successfully Deliver the First 15,000 Ft Lateral Well in an Unconventional Drilling Operation

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226974-MS](https://doi.org/10.2118/226974-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The paper addressed the main technical challenges encountered while drilling the first extended reach 15,000 ft lateral well in an unconventional drilling project. A series of systematic strategies was proposed during the planning phase and precisely implemented during the operation. The success of the well delivery established the best practice that will lay a solid foundation for the future operation to drill the extended reach lateral wells.

For the 15,000 ft lateral wells, limited weight transfer, excessive torque, high pumping pressure, high bottom-hole-circulation-temperatures (BHCT), and the limited drillpipe margin of overpull are the major challenges for drilling and tripping. While casing buckling is the key challenge to deliver the casing to bottom. In order to overcome these challenges, well profile optimization, detailed torque and drag simulations, and optimized mud management procedures were applied. A high operational torque casing string was selected and air-floated to successfully deliver it to the bottom. The optimized well profile helped reduce the torque and drag, enhanced the weight transfer to improve the drilling performance, and assisted in running casing to bottom. The BHA with high temperature tolerance reduced the risk of failure and enabled the well to be drilled to TD in one run. The optimized tripping procedure helped overcome the drill pipe tension limitations at TD. A special mud management system with low rheology and low gravity solids (LGS) optimized the hole cleaning efficiency and surface pump pressure, improving the drilling, tripping and casing running process. Floatation of the production casing, the specially designed centralizers, and an optimized mud system led to reducing the buckling tendency of the casing and reduced the required rotation intervals to run the casing string to bottom successfully.

Key Words: systematic strategies, technical challenges, first 15,000 ft lateral well, unconventional drilling operation

Introduction

Recent efforts to further optimize the value of unconventional assets require innovative approaches that enable lower cost per barrel of developed resources (\$/BOE), to this end, increasing overall lateral length

whereby the upper well sections or overburden remained equal with stretched horizontal sections offer greater reservoir contact at reduced number of wells to develop a defined acreage area.

Unconventional resources development in an area of interest has been underway for quite a few years, and a progressive learning curve and optimization journey have been established to constantly prompt the sophisticated strategy for the drilling and completion group to deal with the tough technical and operational challenges.

From the initial Unconventional wells with a lateral length of 5,000 ft to recent accomplishments of record well with 15,000 ft effective lateral length, the drilling group has been challenged in several aspects. Some of these challenges are associated with the hydraulics and torque and drag (T&D), which had to be evaluated in order to drill these heightened laterals with an efficient and optimized rate of penetration (ROP) that preserved the accrued value of prior designs in 5,000 ft and 10,000 ft horizontal lengths, respectively. The rig fleet in the said project has to adapt and overcome these technical challenges in order to safely deliver these record long lateral wells. Moreover, the rigs ought to be able to safely retrieve the BHAs at TD without exceeding the design limits of the available drill string; these are just a few of those challenges that were worked out.

Another very important consideration for such a stretched horizontal well is to have the ability to safely and successfully deliver the production casing to bottom. In 2017, a mono-bore well design was introduced, and it marked a significant shift, allowing for the drilling of a larger size hole and the running of 5-1/2 in casings all the way to the surface. This new design substantially enhanced the ROP and reduced well delivery time; nevertheless, as lateral length approached 10,000 ft, casing delivery to the bottom of the well once again became a challenge as measured friction factors in open hole in many areas ranged between 0.4 to 0.5, which made the casing runs difficult as the moderate to severe casing buckling was triggered constantly, it became clear that a new approach is required to successfully deliver the unconventional wells with the extending lateral lengths.

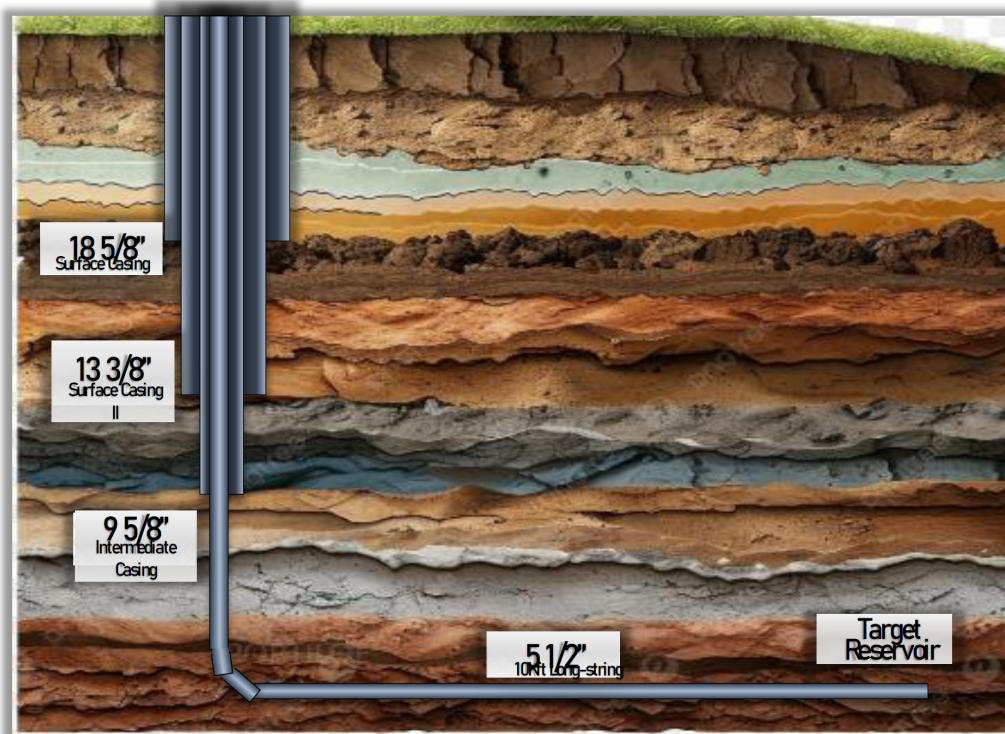


Figure 1—Schematic for a Typical Mono-Bore Unconventional Well with 10,000 ft Horizontal Length

Technical challenges for drilling a 15,000 ft lateral well

As the lateral lengths are extended from 10,000 ft to 15,000 ft, a multitude of complexities arise, posing significant threats to operational efficiency and drilling success. A thorough examination of the operational sequence reveals several critical challenges that are likely to emerge, including those related to drilling, completion, and overall well performance. These anticipated hurdles underscore the need for careful planning, strategic risk management, and innovative solutions to ensure the successful execution of the extended-reach well.

Drilling Hydraulics with Existing Rig Fleet

Drilling hydraulics and its optimization had played a critical role in achieving efficient and safe drilling operations in extended reach drilling applications in the area of interest to this study; Unconventional resources wells with lateral lengths of 10,000 ft and a MD/TVD ratio of 2.1 require optimized drilling hydraulics to ensure effective drilling and casing run operations with zero un-scheduled events. As it pertains to horizontal drilling, by carefully managing variables such as drilling flow rate, drill-string configuration, nozzle diameter, and mud properties, drilling engineers can optimize penetration rates while ensuring constraints like hole cleaning, bit specific energy, and system pressure distribution are satisfied. Hydraulics optimization not only enhances drilling efficiency but also reduces costs associated with the prolonged drilling time caused by potential hole cleaning, wellbore instability, and other issues. Additionally, it ensures that drilling operations remain within the capabilities of the rig and adhere to safety and environmental standards. In the Unconventional project studied in this paper, some constraints on equipment ratings pertaining the existing rig fleet, rotary steerable system, and available resources were assessed to evaluate the feasibility of extending the lateral lengths to 15,000 ft.

Drilling Torque

The ROP in the area of this study is very dependent on surface RPM, predominantly in combination with adequate weight transfer throughout the entire curvature and horizontal section to the bit. During the drilling process, the drilling torque is mostly dependent on the configuration of the BHA, the wellbore profile and hole geometry, the WOB, and the type of formation to be drilled. According to Robello et al. [1], the total surface rotary torque for drilling operations is comprised of two elements:

$$T = T_f + T_s \quad \text{Equation 1}$$

Whereby:

T_f : the torque at the face of the bit, ft-lb;

T_s : the torque component due to the side force F_s , ft-lb;

As far as the bit face torque, for a drag bit, torque can be calculated by modeling it as a simple circular shaft, with a flat-bottom, as follows:

$$T_f = \frac{1,000 \times \mu \times WOB \times Db}{36} \quad \text{Equation 2}$$

Where,

μ : friction factor

WOB: weight on bit, klbs

Db: Diameter of bit, inch

The side force component for a parabolic distribution can be ascertained from the following relationship:

$$T_s = \frac{4}{3} F_s \times \mu \times Rb \quad \text{Equation 3}$$

Where,

F_s: the side force against the drill string, klbs

R_b: radius of the drilling string, inch

Drilling in the subject Unconventional field has required high rotary speed (160-180 rpm) and weight-on-bit (WOB) of 40-60 Klbs to maximize the ROP, a challenge arises when extending the lateral lengths to 15,000 ft from traditional 10,000 ft for the output of the effective torque with the existing top drive system (TDS). As shown in Figure 2 below, maximizing the ROP will have to be planned in accordance with available surface Top Drive Equipment Specs, and best possible BHA design.

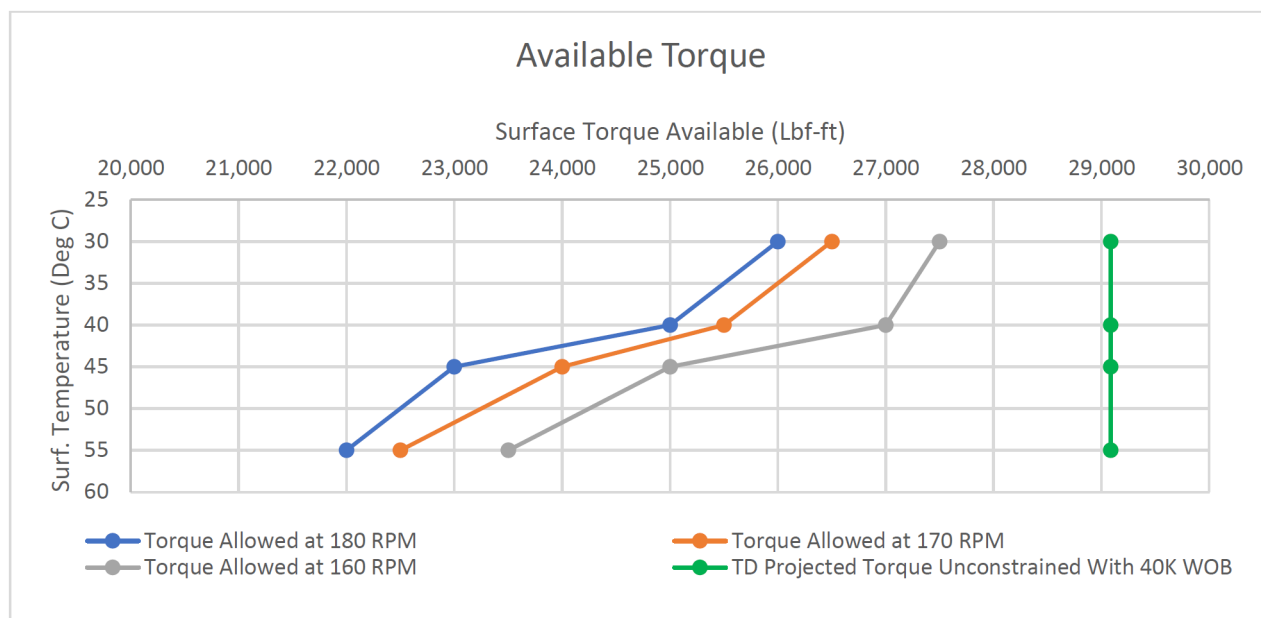


Figure 2—Surface Torque Available vs. Surface Temperature and RPM

Weight transfer issues due to the high drag and Buckling Management at excessive horizontal lengths

Axial Drag can become a hindrance in extended reach (ERD) wells. As the wells get deeper with higher inclinations and more complex hole geometry, drag is commonly considered an important factor to impact the slide mode drilling, however, even in continuous rotation mode with rotary steerable systems (RSS), it is also very challenging for the well profile and BHA design to achieve the optimum weight transfer to the bit through the drilling string. The high shocks and vibration, especially the stick-slips, while drilling the transition zone between soft-hard formations, and also the helical buckling in the rotation mode with excessive slack-off applied on surface, would significantly impact the weight transfer.

During the planning phase of this study, elements such as well path selection for drilling and casing run, drill string design, and drilling fluids properties were all checked in detail so as to minimize the excessive buckling issues, whether in drilling or casing running.

As lateral lengths increased from 10,000 ft to 15,000 ft, due to the required doglegs for a pseudo vertical section with negative displacement and the curve section, buckling may be initiated below the 9-5/8 in casing shoe, as shown in Figure 3. As per the previous study from Jiang Wu [2], the buckling length and the average buckling load for the drilling string in an inclined wellbore with the combined axial load and torsional load are as follows:

$$L_h = \sqrt[4]{\frac{(8\pi^4 r EI - 2\pi^3 r T_n L_h)}{wsin\theta}} \quad \text{Equation 4}$$

$$F_b = \sqrt{\frac{\left(8EI - \frac{2T_n L_h}{\pi}\right) w \sin \theta}{r}} \quad \text{Equation 5}$$

Where,

E: Young's modulus, psi

I : moment of inertia of tubulars, in.⁴

Fh: average helical buckling load, lbf

Lh: length of helically buckled pipe, in.

R: radial clearance between wellbore and tubular, in.

Tn: torsional load, lb-in.

W: tubular weight in mud, lb/in.

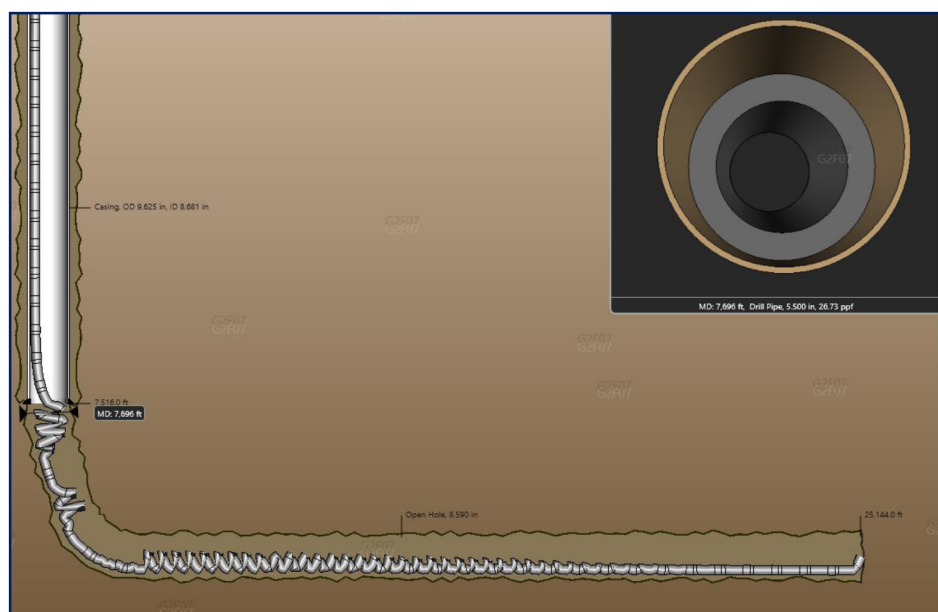
 θ : wellbore inclination, deg.

Figure 3—Buckling Profile Rotating on Bottom

The simulation revealed the potential for buckling located between the 9-5/8" casing shoe and the kick off point (KOP) and throughout the curve section. This physical behavior results from multiple factors such as the well path tortuosity, hole geometry, insufficient pipe stiffness, inadequate hole cleaning, etc.

With the severity of buckling increasing, the initial sinusoidal buckling (or lateral buckling) may develop into helical buckling (or spiral buckling), for the worst case, it may get into the lock up (or post-buckling deformation), in which the pipe can't move down any, to some extent, it is as mis-judged as pipe stuck.

From the modelling chart above for Torque and Drag (T&D) Analysis with the drilling parameters contemplated in the standard well road map (WOB/RPM: 30-65 Klbs / 160-180 rpm), there is a small margin in terms of compression that might trigger Helical buckling of approximately 20 Klbs, furthermore, increasing expected measure depths to 27,000 ft MD might rendered this small margin almost to zero, this was a challenge that needed to be looked at from an engineering point of view to ensure a successful 15,000 ft lateral pilot test.

Challenge for Drilling Fluids Management

During the execution of prior 10,000 ft lateral wells, drilling fluids have played a key role to ensure a stable wellbore with proper filter cake across permeable formation, adequate hole cleaning, and appropriate ECD management. Extending the horizontal lengths of the subject field wells to a total measure depth (TMD) of

26,800 ft from a previous 21,600 ft MD, which stretched the open hole section up to 19,350 ft. This change would bring many changes for the drilling fluids management in terms of weight, rheology, solids control, and sweeps pumping to enhance hole cleaning, reduce friction so as to successfully drill to TD of the well, and also deliver the casing to bottom.

Downhole directional tool failure due to the high BHCT

High bottom hole circulation temperature (BHCT) is challenging the tool functionality. The bottom hole static temperature (BHST) for the subjected area of interest can be up to 260 °F. The directional tool previously used to drill a 10,000 ft lateral well can only tolerate a BHCT limit of +/-300 °F; however, the BHCT in the operation may be up to 320°F. Once the downhole temperature approached the tool temperature limit, some sensors inside the tool may be out of order, the electrical circuit may lose the communication to surface, and many RSS or MWD tool failures have occurred previously due to the high temperature. The tool had to be tripped out to replace it, but the replaced tool is facing the same risk of failure. Therefore, how to handle the high BHCT to allow the directional tool to be functional properly is also a challenge when drilling the ERD well.

Production Casing Run to TD with Minimized Buckling

Since the beginning of the Unconventional boom in the USA back in the early 2,000's, well complexity due to heightened horizontal lengths had brought forward substantial hurdles for running production casing strings in ever increasing lateral footage per well; in the subject field of this study, one of significant concerns was the 5-1/2 in long production casing run in 8-1/2 in hole. During the well construction process with a pre-determined narrow well placement target window, a well trajectory that is followed and geo- steered brings doglegs well above 2 deg per 100 ft which coupled with fast ROP's (Average of 120 ft/hr including survey and connections) posed a challenge to ensure proper hole cleaning prior to running the production casing. This study expands on this reality as it identifies an approach that will facilitate the casing run operation with enough hook load available and minimized buckling.

The historical drilling of most 10,000 ft horizontal lateral wells in the subject field had indicated a cased hole friction factor (CHFF) ranging from 0.15 to 0.20 as well as an openhole friction factor (OHFF) ranging from 0.4 to 0.5. The OHFF is considered very high, which would severely diminish the ability of our execution teams to run production casing strings to planned TD with 15,000 ft horizontal measure depths without significant buckling loads. [Figure 4](#) below shows an example of a typical casing run situation for a 10,000 ft horizontal lateral well. As can be seen, the actual hook loads indicated that the OHFF fell in a range between 0.35 to 0.50.

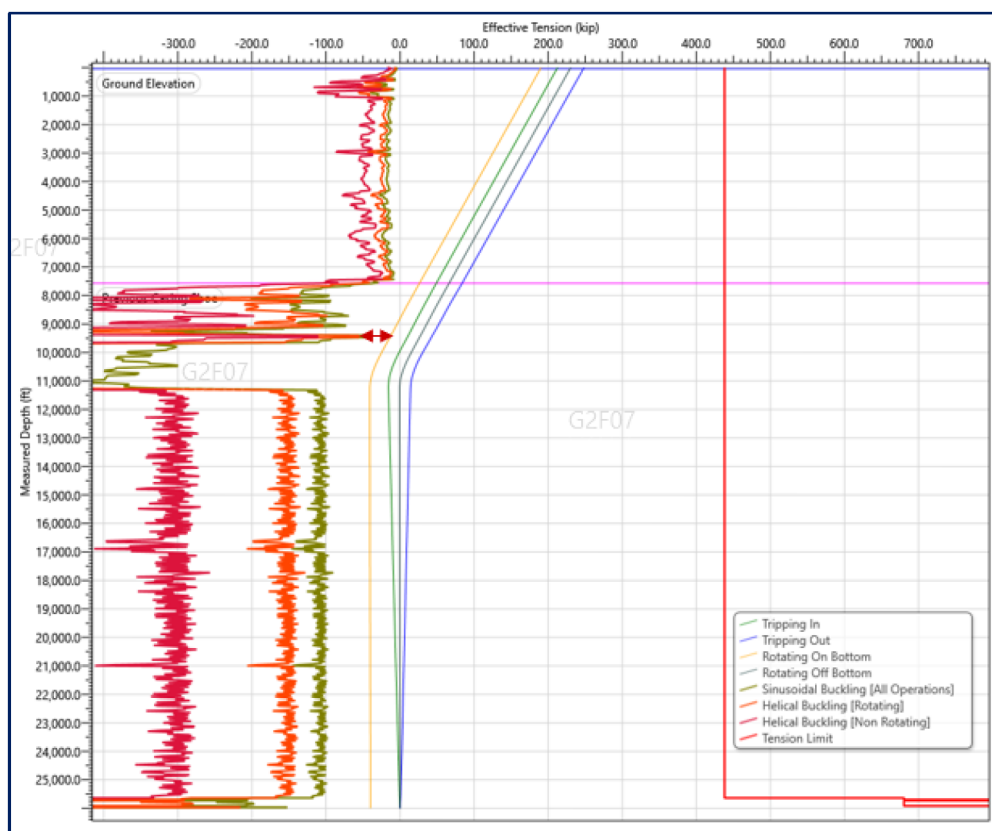


Figure 4—Effective Tension Modeling with BHA for standard 10,000 ft Horizontal Wells

Previous study from G. Yin, et al [3] indicated that for the horizontal wells with more than 10,000 ft lateral, the flotation is a necessity to successfully deliver the casing to bottom. However, with the lateral excessively extended, how to ensure the shoetrack is fully functional all the time and the rupture disk of the flotation device is undamaged throughout the casing running process is another challenge.

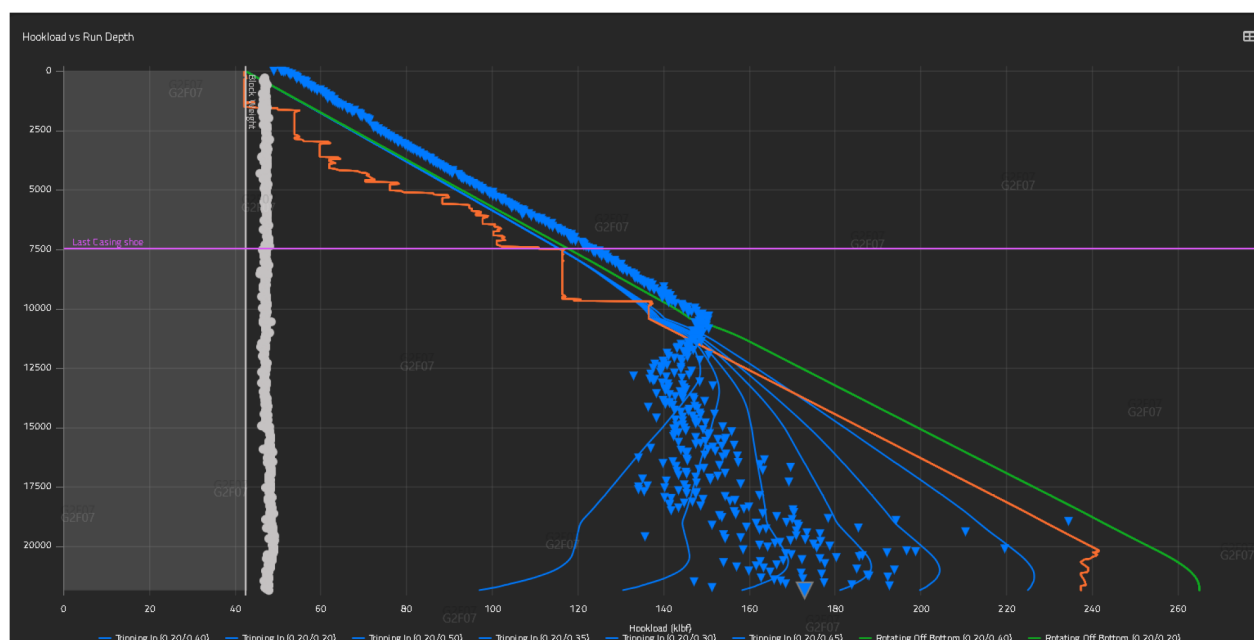


Figure 5—Typical T&D Profile for a Horizontal well with 10,000 ft Lateral (OHFF: 0.35 to 0.5)

Design and Engineering Approach for 15,000 ft Lateral Wells

Drilling Hydraulics

One of the first variables analyzed from an engineering standpoint was the capability of the existing rig fleet to manage the required minimum flow rate and stand pipe pressures to effectively clean the wellbore, per engineering modeling with several flow rate scenarios, it revealed that at a flowrate of 560 GPM the surface anticipated pressure would reach 5,180 psi which is still within the operating envelope of the existing pumps dressed with smaller liner size. A hole cleaning and cuttings bed modeling revealed that at a flowrate of 520 GPM with an instantaneous ROP of 350 ft/hr, there would be very minimum cuttings bed formed along the wellbore, if the instantaneous ROP is increased to 420 ft/hr and the rotary speed is reduced to 120 RPM, there would be 0.09 inches of cuttings bed formed in the wellbore (Figures 6 and 7) throughout the lateral.

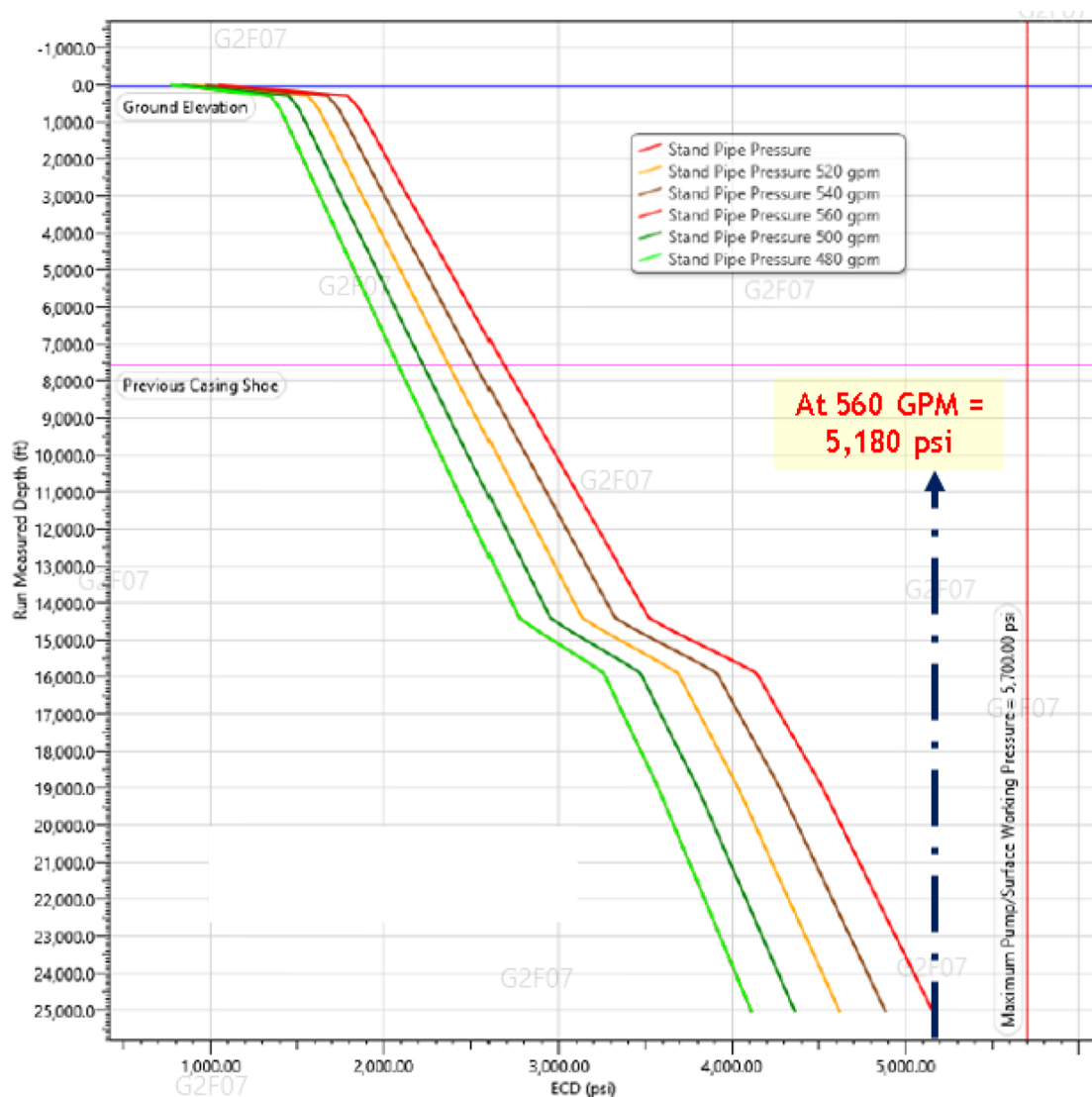


Figure 6—Surface Pump Pressure Profile at Various Flow Rates (GPM)

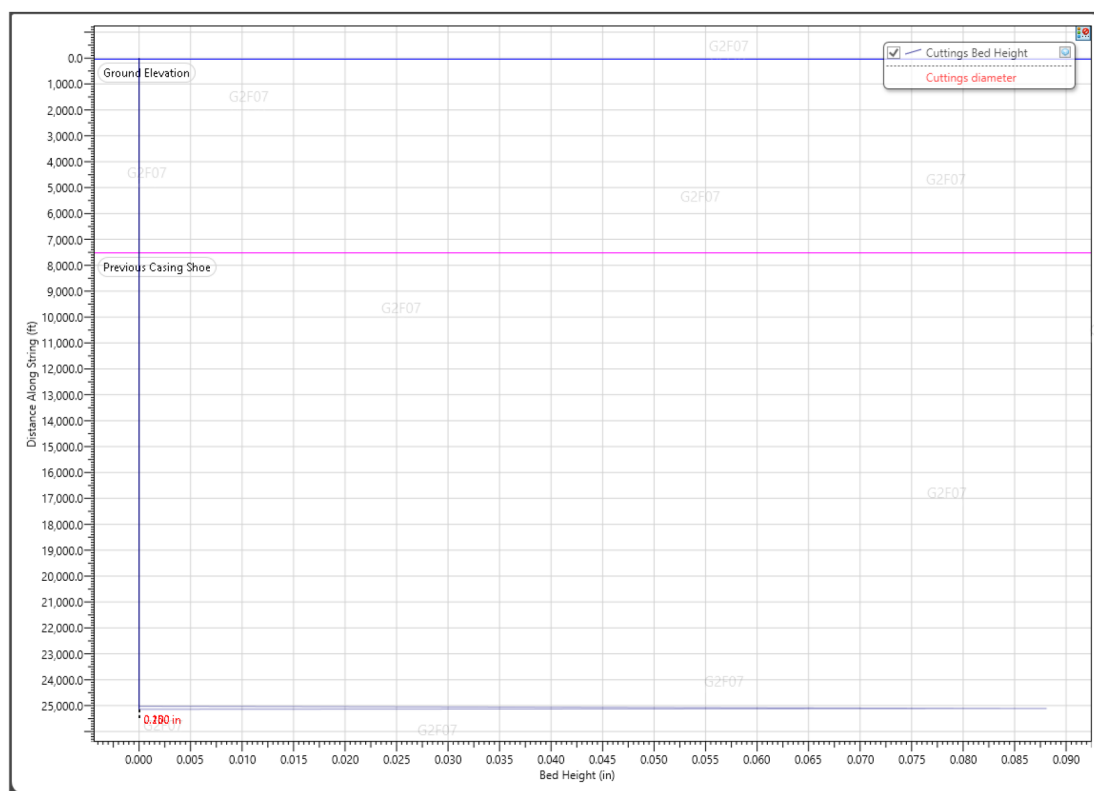


Figure 7—Cuttings Bed Height vs. Depth, Drilling With 520 GPM

The parameters used in the above model for cuttings bed development were:

Parameter	Magnitude
Drilling Flow Rate (GPM)	520
Cuttings Diameter (inches)	0.1 to 0.2
Cuttings Density (S.G)	2.67
ROP (ft/hr)	350
Rotary Speed (RPM)	150

Drilling hydraulics modeling also showed that even at a lower flow rate, such as 480 GPM with a lower instantaneous ROPs below 200 ft/hr, it was possible to maintain good hole cleaning while keeping a minimum of 140 RPM.

Drilling Torque

Standard for 10,000 ft Horizontal wells in the field subject to this study aims at optimizing ROP while delivering a quality wellbore that facilitates subsequent casing runs to total well depth. As 15,000 ft horizontal wells became an aspiration, a new approach for well trajectory was used, which would enable reaching the total horizontal lengths while at the same time permitting the highest probability of success while running casing. Traditionally, the wells in the subject unconventional field were drilled with a straight hole to KOP, then built a curve section to a landing point (LP) with a build rate between 5.5 to 6.5 degrees per 100 ft; with the advent of 15,000 ft lateral wells, a new directional approach (Figure 8) with a back build was adopted which showed better ability to slack off more weight to TD with casing runs but slightly

higher torque if rotation was needed with the casing or drill string depending upon the open hole friction factor reached (OHFF).

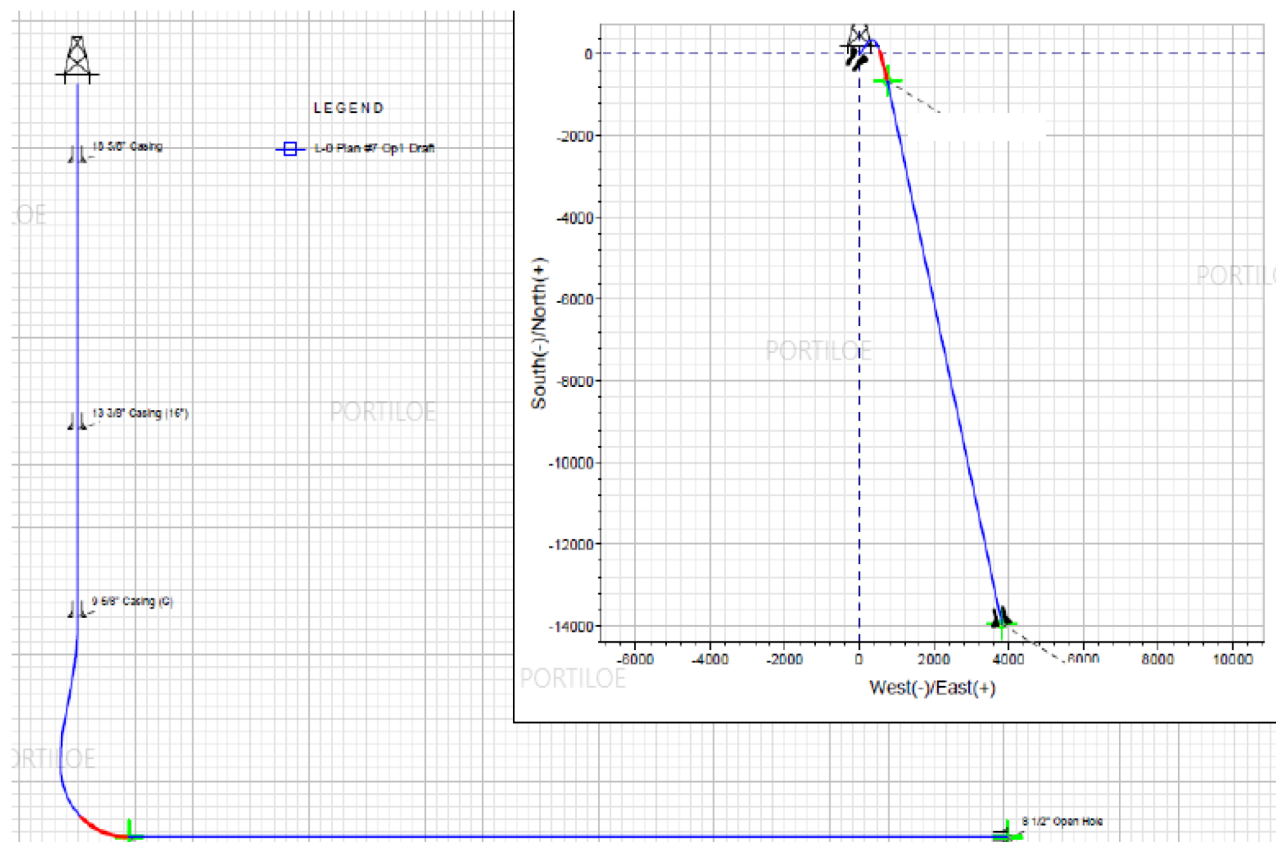


Figure 8—Negative Build Profile Adopted for 15,000 ft Lateral Length Wells

Current unconventional wells with 10,000 ft lateral sections have reached torque values that might range from 26,000 to 31,000 ft-lb at TD. These values are expected to further increase as lateral lengths are extended by an additional 5,000 ft of lateral section.

With actual offset survey and measured tortuosity, one simulation with advanced T&D software was performed to understand the torque requirements to drill a 15,000 ft long lateral well. According to the analysis, it revealed that when drilling with a targeted rotary speed of 180 RPM and WOB 30-50 Klbs, the estimated surface torque value may exceed 30,000 ft-lb, (as shown in Figure 9), which has been beyond the top drive system (TDS) torque output limitation for the assigned rig for this project.

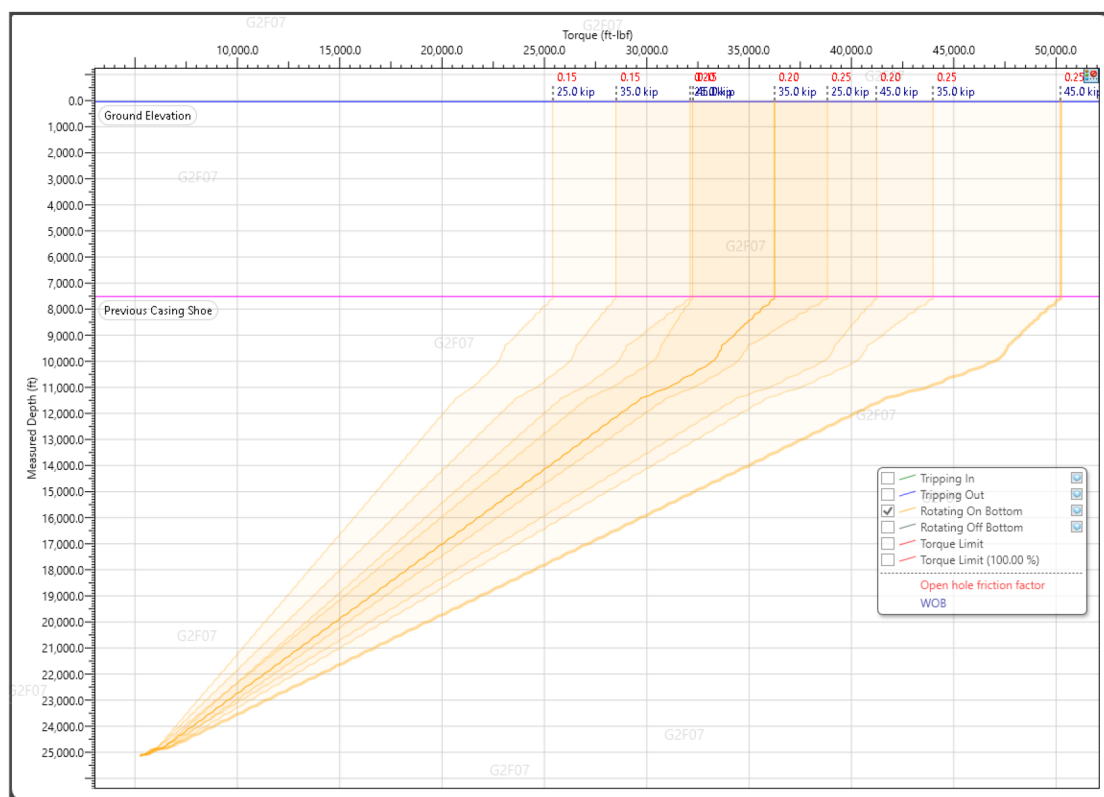


Figure 9—Torque Modeled for 15,000 ft Laterals with TQFF range of 0.15-0.25 and WOB of 25-45 Klbs

In order to deal with the new torque requirements for drilling the 15,000 ft long lateral well, a phased approach (Table 2) to step down the rotary speed while drilling the lateral section has been adopted so that the higher torque can be output to meet the drilling requirement.

Table 2—Phased Approach to Step down rotary speed to allow the output of the needed Surface Drilling Torque

Depth Range (ft)	Surface RPM	Surface Allowable Torque (Ft-Lb)	WOB (Klbs)
7,526 to 22,000	180	26,000 to 27,000	30-50
22,000 to 24,000	160	27,500 to 28,500	30-50
24,000 to 26,800	140	30,000 to 31,500	30-50

Figure 10, below, depicts an example of a Torque vs. RPM curve depending upon each specific rig equipment. The intent of this curve is to ensure that drilling efficiency and ROP maximization are commensurate with the capability of each specific drilling unit.

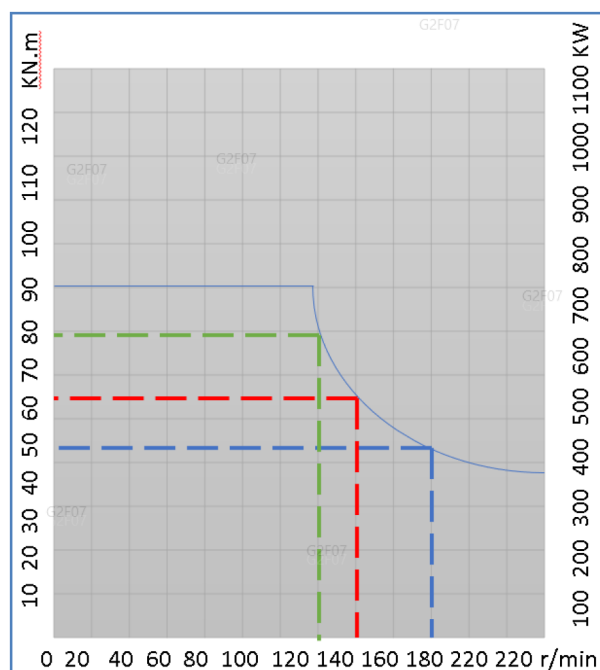


Figure 10—Phase Approach for Torque Requirements at Increased Horizontal Lengths (15,000 ft Laterals)

Drilling Dynamics and Axial Drag

The main goal of 15,000 ft horizontal wells was to maximize reservoir contact. With this new well profile adopted, the well was built toward the opposite direction of the lateral and then steered in the intended direction before the well is landed (Negative Build), the vertical section was reduced by around 400 ft. As a result, the same length of reservoir will be added. As the profile of the well is changed, an engineering aspect to assess is downhole dynamics and axial drag. Calibration from offset data suggested that a TQFF ranged between 0.15 and 0.2; with these boundary conditions, two different BHA approaches and drilling scenarios were assessed: one is the similar BHA used to drill previous 10,000 ft laterals wells; the other one is the BHA with a modification to existing BHA design by adding heavy-weight-drill-pipe as push pipe.

From Figure 11 above, it can be seen that if the drilling teams attempted to maximize ROP, a WOB to a margin of 15-20 Klbs max would be available before Helical buckling occurs.

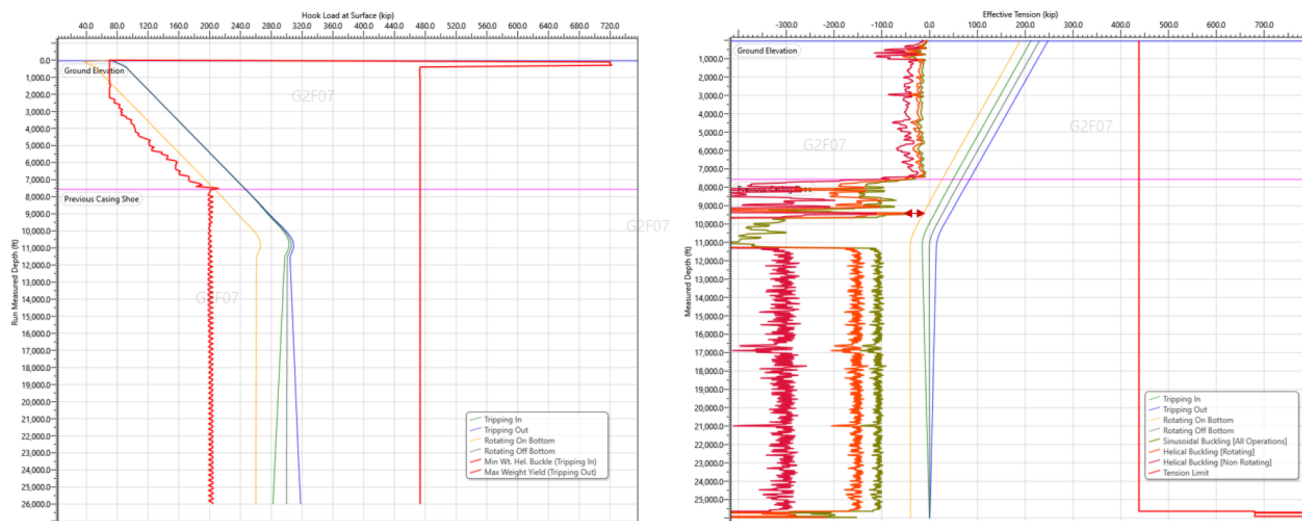


Figure 11—Torque and Drag Modeling Without Push Pipe

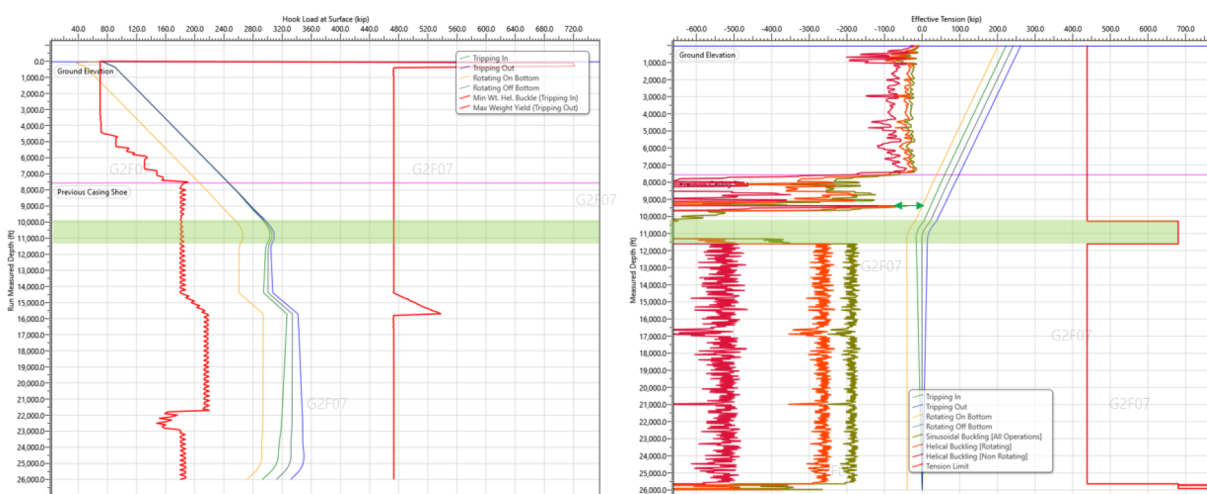


Figure 12—Torque and Drag Modeling With Push Pipe

As drilling conditions and well tortuosity were estimated based on offset wells, it was established that a different approach to BHA design could be trialed for the 15,000 ft lateral wells. A push pipe was added to the drill string, aiming at reducing the pipe buckling from axial loads that would enable more effective weight transfer. This change was a successful factor for the first 15,000 ft lateral well, which was drilled with one BHA shoe-to-shoe and with better than expected ROP ahead of planned days (5% ahead of plan). Figure 12 above shows that with the push pipe implemented, the margin to potential Helical Buckling increases from 15 Klbs to approximately 65 Klbs.

Drilling Fluids Management

Drilling fluids played a key role in any ERD operation, because they would impact many aspects such as hole cleaning, hydraulic pressure, lost circulation associated with ECD management, friction reduction, hole stability, etc. Drilling Fluids are also of utter importance to successfully running casing to bottom with minimized likelihood of casing rotation required. For this trial of extended horizontal lengths, it turned out that a fine-tuned mud formulation was necessary; low gravity solids (LGS) were targeted to be controlled under a certain limit by strict solids control management and dilution schedule. While drilling and running casing, the following best practices were implemented:

- While drilling, stick to the programmed target of LGS as 14% and drop to <10% for the last 4,000 ft before TD.
- Mud samples will be taken (at landing point, mid-way of the lateral, and at TD) and will be sent to the mud facility for analysis of Particle Size Distribution (PSD). This is required to assess the source and amount of LGS.
- Solid control equipment will be working in solid removal mode at all times.
- Ensure an estimate and measure of the rate of barite removal and its effect on density are in place for using solids removal mode only, and determine the rate of barite removal, the mixing rate, and volume required for replacement.
- All necessary measures are taken to maintain density while processing fluids in solids removal mode.
- Stop using wellbore strengthening treatment (bridging material) after the landing point, this is to minimize content of LGS.
- Perform dilution as required.
- Shaker screens will be with 170 mesh or higher after the landing point.

- Percentage of LGS required to have before running casing, in the range of 7-8%.
- Fann readings 6/3 rpm during clean-up cycle before tripping out should be at 8/7. A drill fluid system with the same density reached at well TD must be ready to be displaced into the wellbore to meet all the LGS and rheological requirements prior to running casing, this will ensure an optimum formulation is left in the hole that will help facilitate the casing run and contribute to achieving open hole friction factors less than 0.30.
- Perform dilution as required if backreaming out of the hole at TD is needed.
- Verify and report the efficiency of Solid Control Equipment (SCE) before starting drilling the 8-1/2" section (screen of shakers, efficiency of desander and desilter). Take the necessary actions to increase the efficiency of Solids Control Equipment.

Alternatives to handle the high BHCT issues

During the planning phase, a high temperature tolerant directional tool (RSS and MWD) is selected to use. According to industry knowledge, the major impact factor for the downhole circulation temperature is the flow rate, and the rotary speed will be secondary. As such, a detailed drilling roadmap associated with flowrate, rpm, and WOB control was well established.

Engineering Approach to Casing Running for 15,000 ft Lateral Wells

Increasing the length of the 5-1/2 in Production Casing string by an additional 5,000 ft required a comprehensive approach that sought to deliver the best possible wellbore conditions to minimize the open hole friction factor (OHFF) from traditional 0.40 to 0.45 as seen in 10,000 ft lateral wells to equal or less than 0.32. In addition, a contingency plan to have a casing string with the ability to rotate while being floated was to be implemented. A dedicated casing with premium connections equipped with sufficient available operation torque was selected for the first trial of the project. Some of the design and engineering considerations planned are detailed below:

a. Drilling fluid system:

The 15,000 ft lateral wells have been planned with two fluid systems, one to drill with a 90/10 OWR and minimized LGS (11-13%) which enables to drill the wellbore with sufficient mud rheology to clean the wellbore along its entire length, rheological properties were constantly monitored and adjusted via Dilution and full utilization of the solids control equipment.

b. Hole cleaning prior to running Casing:

At well TD, a fresh mud system with very low LGS was used to displace existing mud after the first two clean up cycles. This system had an even improved rheological profile, which included strict control of LGS between 7-8%. Subsequently, during the tripping process, backreaming the first 8,000 ft of lateral from TD. This was to be done due to the drillpipe tension limit. During this process, the cuttings from the wellbore would be further removed. Afterwards, another set of clean up cycles would be performed with the weighted sweeps pumped so as to ultimately ensure the cleanest possible wellbore condition prior to running the production casing.

c. Selection of the casing and the proper running procedure to prepare for the contingent rotation required:

A casing with premium connections with high operating torque limits was selected. Utilize the Jam Unit System with AI algorithm to qualify each connection to be made up to a torque of 200 ft-lb below the maximum make-up torque. The 15,000 ft lateral wells presented some uncertainty, on one end, if OHFF remained low as planned between 0.25 to 0.30, then the long production string could be run to bottom without need to rotate, nonetheless, if OHFF spanned from 0.32 for open hole or above, then some rotation might be needed, in which case, a careful usage of premium connections allows to rotate the casing at 20-30 rpm to its maximum operating limit which would help deliver the casing to bottom.

d. Careful selection of guide Shoe and flotation device:

To deliver the extremely long production casing to the bottom, one of the most important considerations is to use flotation technology. To ensure the flotation system is not compromised during the casing run, two failures must be avoided. One is the breaching of the shoe track, in which, the float valve is broken and the mud flows back to the inside of casing from the annulus, causing the failure of the flotation. The other is the premature breaking of the rupture disk for the flotation device. Meanwhile, to minimize the friction during the casing run, the guide shoe should be small in size and adept to rotation requirements while it is strong enough to tolerate any collision and shocks damage. To meet the requirement above, a dedicated guide shoe with an eccentric nose design, slick, and smallest diameter, and 15,000 psi backpressure was used. A flotation device with 10,000 psi resistance pressure from top and bottom sides was adopted. The unique guide shoe selected allows for the best chance to slack off to TD, for all scenarios modeled while slacking off and/or rotating the production casing to TD. When blades are introduced to the casing shoe, one must make sure they are small in diameter and straight blades (not tightly wrapped). If residual cuttings beds are still in the wellbore, tightly wrapped blades will not allow cuttings to pass through the junk slot.

e. Casing Centralization:

For 15,000 ft lateral wells, the same centralization setting was used as that in 10,000 ft lateral wells, in which the centralizers were optimized to install to maximize the chance of delivering the casing to the bottom successfully. One distinction was the use of a different type of centralizer that allowed at least a 10% to 15% friction reduction when compared with other alternatives. As per modeling, this provides a better scenario to run the production casing string without rotation.

f. Shoe Track Configuration:

With a consideration to minimize friction and enhance casing running purely on slack off weight and without rotation, a new approach to conventional 10,000 ft laterals was used in which the first centralizer was placed right above the shoe and fixed, upward, no centralizer to be installed until the 2nd centralizer placed at the middle of the 3rd casing joint from the shoe. As such, the distance between the first and 2nd centralizer is approximately 80 ft. This approach aimed at providing a "Nose Up" profile, which ultimately facilitates the casing run through some wellbore dog legs along its entire profile, [Figure 13](#) illustrates this shoe track configuration:

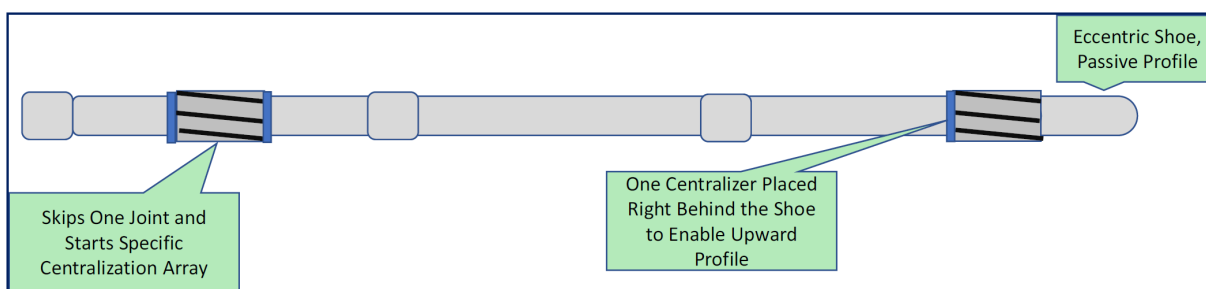


Figure 13—Shoe Track Configuration for 15,000 ft Lateral Wells

Execution of Drilling and Completions of the 1st 15,000 ft Lateral Well

The first 15,000 ft horizontal lateral well was successfully drilled to a measured depth (MD) of 26,000 ft at a true vertical depth (TVD) of 10,620 ft. The 8½" section was drilled in one single run, starting from the 9⅝ inch casing shoe until the well TD without any tripping in between, covering a total length of 18,441 ft of 8-1/2" open hole. The BHA was smoothly tripped out of hole without any major events or any stuck pipe incident, and the 5½" production casing was successfully floated to the target setting depth of 25,934 ft

without any casing rotation. However, several Extended Reach Drilling (ERD) challenges were encountered during the process of tripping out of the hole and running the 5½" casing.

Figure 14 below shows the as-built well profile used. The negative built well profile enabled a higher hook load profile, providing sufficient weight to run the casing to bottom without any casing rotation. The casing was run with an assisted casing flotation device that was installed at 45 deg inclination.

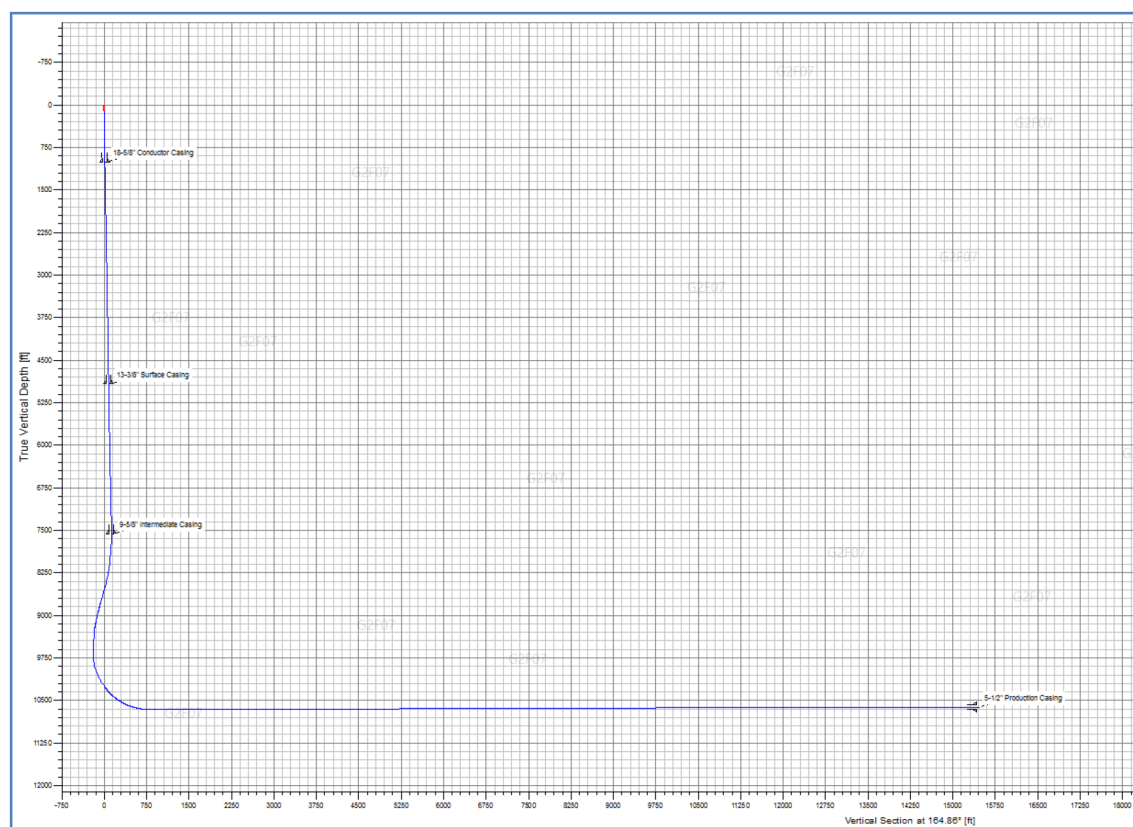


Figure 14—As Built Well Profile with Negative Build for Higher Non-Rotating Hook Load

The well archetype used for this 15,000 ft lateral well was based on the traditional 10,000 ft well design (Figure 15) which consists of two surface strings 18-5/8 inch and 13-3/8 inch to isolate unconsolidated formations and aquifers present in the basin, as well as an intermediate 9-5/8 inch casing set deep enough to isolate lower pressure and permeable formations. A blow-out preventer (BOP) is installed and pressure tested after 13-3/8 inch casing was run and cemented, a formation integrity test (FIT) was performed below the 9-5/8 inch casing shoe to assure the formation integrity to be high enough to reach the target depths.

At well TD, six bottoms-up circulations were performed with maximum flowrate and rotary speed applied, and the weighted sweeps were pumped as needed. The first four bottom-ups were conducted with existing mud, while the remaining two were done to displace and condition a new, fresh drilling mud due to a high concentration of low-gravity-solids (LGS), which reached approximately 15% (Target Max 13%). After circulation was completed, rig backreamed out of the first half of the lateral section (~7,000 ft) to a depth around 19,000 ft. While doing the backreaming, some instances of high torque and partial packing off tendencies occurred because of the high backreaming speed. Although there was one string stall accompanied by a pack-off of about 500 psi, no stuck pipe events occurred during back-reaming. After circulating the hole clean again at 19,000 ft with three clean up cycles, the remaining 7,500 ft of the lateral and the curve section were tripped out on elevators without major issues.

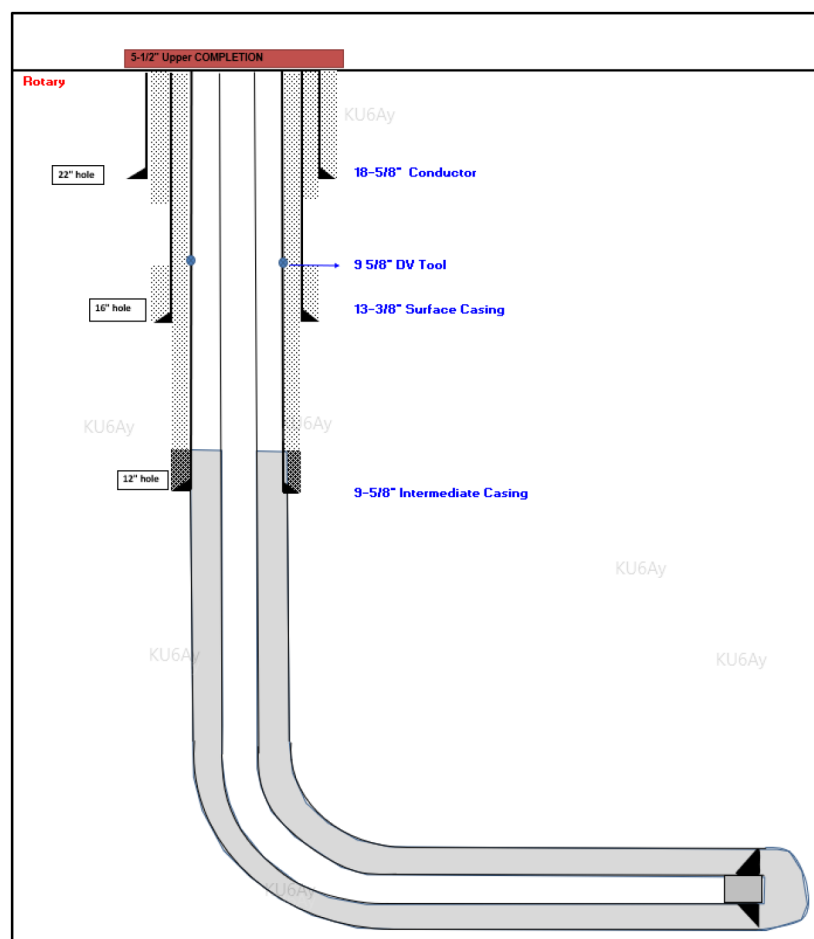


Figure 15—Typical Well Archetype

Based on offset analysis, three formations in the vertical section of the well are highly permeable, leading to excessive filter cake build-up due to the high LGS content. This also increased the risk of differential sticking. As a precaution, the vertical section was back-reamed, specially across transition zones. While some erratic torque was observed during the back-reaming of the vertical section, it was not accompanied by packing off, and the BHA was successfully pulled to the surface without any major problems.

The 5½" long string of casing was tripped in empty to a depth of approximately 15,350 ft, whereby the casing flotation device was picked up and run in, filling the pipe every five joints to TD. There were no major hole condition-related restrictions during the casing run, except at around 22,800 ft, which corresponded to a transition zone (as indicated on the gamma ray log) and required minor working to pass through. The casing was successfully run to the target setting depth of 25,934 ft without rotation, with a final slack-off OHFF of approximately 0.27 at TD. To assist in planning future wells, three joints were purposely rotated near TD, from 25,506 ft to 25,630 ft, to evaluate the torque friction factor and the positive impact of rotation on the slack-off hook-load. As the casing was run through the lateral, hook-load fluctuations increased, indicating that buckling occurred, which was mitigated by slowing the running speed or picking up the string into tension to release the buckling. Rotation off-bottom torque measurements were taken during the casing run to help calibrate the T&D model. Additionally, four successful slick-line drifts were performed as a precautionary measure to verify the casing was not deformed due to the bellied out connection, so as to ensure the drift internal diameter (ID) met the required size to allow the plugs to pass in the subsequent operation sequences.

At the target depth, the casing flotation device was burst by pressuring up to approximately 2,740 psi, and the casing was filled with mud. Then the trapped air was removed, the circulation was established

with no issues. No H₂S and other gases were noted after one bottom up. The cement job was conducted successfully afterwards. Overall, the entire completion phase of work, including casing run and cementing, was an operational success from the beginning to the end.

Figure 16 above shows the torque and drag modeling vs. actual hook loads. It can be seen that the OHFF at first trended along 0.30 and then dropped to 0.27 as the empty casing string was pushed down by the upper casing above the flotation device that was filled with mud. This phase of the casing run was of utter importance to finally deliver the entirety of the expected reservoir net pay.

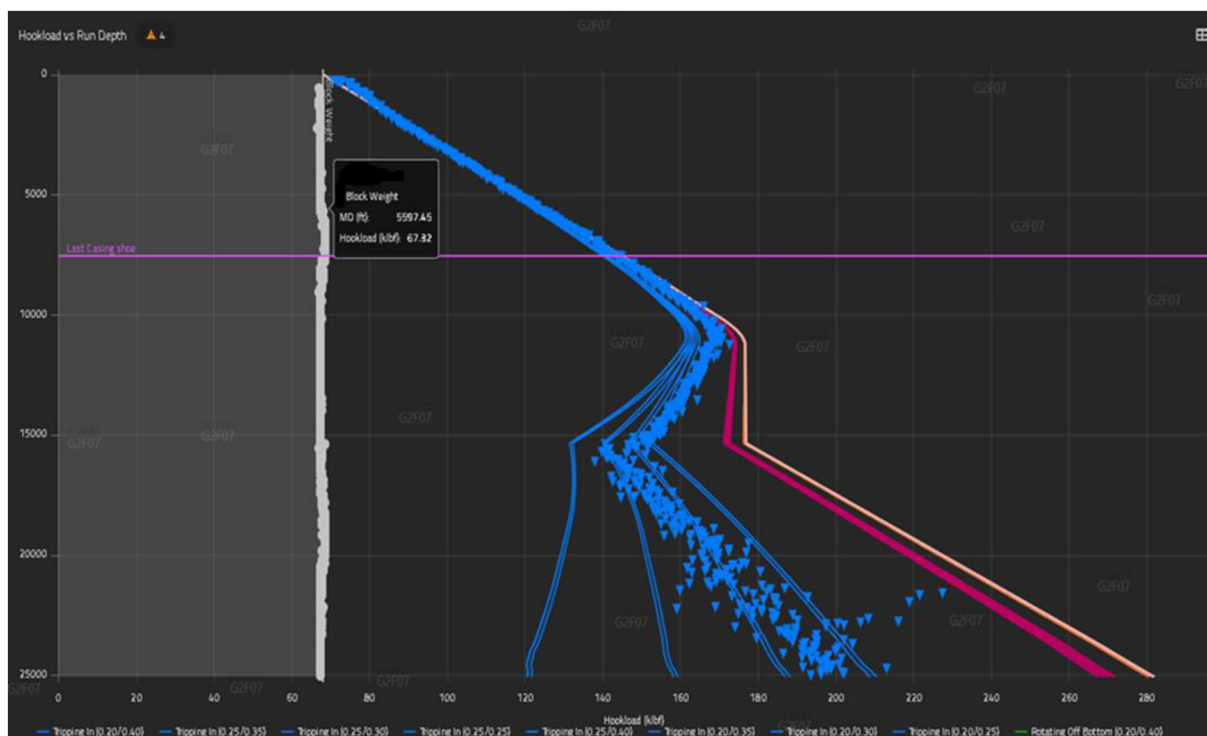


Figure 16—5-1/2" Production Casing Slack Off Hook Load Modeling Along with Actual Real Time Data

Conclusions

With the precise implementation of the established strategies proposed in this paper, the first 15,000 ft lateral Gas well was successfully drilled and completed with outstanding performance. The ability to deliver this 15,000 ft lateral well in the operator's field was verified, which marked a new milestone in history and would allow more field development enhancements in the future. Throughout this super lateral well delivery process, the following key conclusions can be drawn:

1. The engineering approach to adopt a negative build profile coupled with a tuned-up mud system displaced at well TD, helped reduce the OHFF to a range of 0.27-0.30 from 0.35-0.50 noted from previous 10,000 ft lateral well drilling, this facilitated a smooth production casing run to total depth without the need for casing rotation.
2. All aspects of front-end planning came as expected, leading to the success of this project. The torque management with the surface RPM phased approach enabled drilling to well TD within the TDS operating torque envelope. In addition, pump pressure behaved well within the ratings of the pump's liners and system pressure, reaching TD with approximately 4,700 psi.
3. Given the performance observed in this 1st ever 15,000 ft lateral well, the rig has shown high potential to drill longer laterals to further improve the production, provided proper upfront planning and equipment are utilized. Enhancements such as using stronger drill pipe at the surface for additional

margin-of-overpull, upgrading to higher torque premium casing connections to allow the casing rotation with no major concern, following a well-planned drilling roadmap, and adhering to the best operation practices are the key steps to achieving more success.

4. The systematic application of all the main elements of design and engineering described throughout this paper as well as consistent and flawless operational execution, were the critical success factors to deliver this major breakthrough in unconventional drilling while being 5% ahead of planned target days.
5. If backreaming out of the lateral is required, it is crucial to execute this process correctly to avoid serious stuck pipe events. Essential steps include circulating clean at TD with as many cycles and sweeps as required, using optimal parameters (high rotary speed and full flow rate), allowing torque, pump pressure, and flow back to guide the backreaming speed, and maintaining patience. Torque spikes during backreaming are often caused by the top of the BHA catching up with the residual cuttings bed being pushed up the hole as the string is backreamed out.
6. It is vital to resist the urge to pump out in a highly deviated wellbore. Rotation plays a critical role in effectively transferring cuttings out of the lateral during clean-up circulation and in cleaning the wellbore above the BHA while Backreaming. Without rotation, cuttings can easily accumulate around the BHA as it is pulled out of the hole, leading to a high likelihood of packing off.

Nomenclature

BHCT = Bottom hole circulation temperature, deg. F

BOE = Barrel of Oil Equivalent

BHA = Bottom hole assembly

TD = Total depth, ft

MD = Measured depth, ft

TVD = Total vertical depth, ft

TDS = Top drive system

RPM = Revolutions per minute

WOB = Weight on bit, klbs

ROP = Rate of penetrate, ft/hr

ERD = Extended Reach Drilling

KOP = Kick off point

ECD = Equivalent circulation density, PCF

TMD = Total measured depth, ft

T&D = Torque and drag

CHFF = Cased hole friction factor

OHFF = Open hole friction factor

GPM = Gallon per minute, unit of flowrate

S.G. = Specific Gravity

TQFF = Torque friction factor

PSD = Partical size distribution

BHST = Bottom hole static temperature

LP = Landing Point

LGS = Low gravity solids

FIT = Formaton integrity test

ID = Internal diameter, inch

SCE = Solids control equipment

OWR = Oil water ratio

BOP = Blow out prevent

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